

Nightmares Course: Sepsis

Section 1: Case Summary

Scenario Title:	Sepsis/Cholangitis
Keywords:	Sepsis, ward medicine, surgical care
Brief Description of Case:	This case involves the approach to severe sepsis, more specifically acute cholangitis. If treated aggressively (IV fluids, early broad spectrum antibiotics and source control) the patient will stabilize. If not the patient will deteriorate into a PEA arrest.

Goals and Objectives	
Educational Goal:	Demonstrate an evidence-based approach to early sepsis management
Objectives: (Medical and CRM)	1. Recognize sepsis in an unwell patient 2. Prioritize initial therapies (fluid resuscitation and early antibiotics) 3. Understand the importance of urgent source control 4. Communicate the clinical situation clearly to consultants
EPAs Assessed:	

Learners, Setting and Personnel					
Target Learners:	☒ Junior Learners		☐ Senior Learners		☐ Staff
	☒ Physicians	☐ Nurses	☐ RTs	☐ Inter-professional	
	☐ Other Learners:				
Location:	☒ Sim Lab		☐ In Situ		☐ Other:
Recommended Number of Facilitators:	Instructors: 1				
	Confederates: 1 (RN)				
	Sim Techs: 1				

Scenario Development	
Date of Development:	2015
Scenario Developer(s):	Dr. Tim Chaplin
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Revised By:	Dr. Chris Heyd
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Deborah Jenkins		Age: 47		Gender: F	Weight: 90kg
Presenting complaint: Confusion					
Temp: 38.9	HR: 122	BP: 92/50	RR: 22	O ₂ Sat: 92%	FiO ₂ : RA
Cap glucose: 5.8			GCS: 14 (E4 V4 M6)		
Triage note: You are covering an inpatient surgical floor. Its 2300 and you are called to assess a 47M who has been admitted for cholecystitis and is awaiting a cholecystectomy. You are called for new confusion and fever that has developed over the last 3 hours.					
Allergies: NKDA					
Past Medical History: HTN			Current Medications: HCTZ		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, confederate, SP, etc.)?</i> Patient is confused. She will complain of abdominal pain but give no other meaningful history. Not oriented to person, place or date.	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Tachycardia	Neuro: Confused, no focal deficits
Resp: Nil	Head & Neck: Nil
Abdo: RUQ tender with guarding, Murphy positive	MSK/skin: Diaphoretic
Other: Nil	

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Critical care/airway/ACLS equipment
C. Required Medications
IV fluids, antibiotics
D. Moulage
Diaphoresis
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> Confusion RUQ tenderness

Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Ward Nurse	The ward nurse present in the room should have a headset to allow communication with the control room. They will be the nurse that called the resident to assess the patient and will be familiar with their past medical history and medications. Nurse will be helpful and non-obstructive.

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus tach HR: 122 BP: 92/50 RR: 22 O ₂ SAT:92 % T: 38.9°C GCS: 14	Confused RUQ tender	<u>Expected Learner Actions</u> ☐ Focused Hx and Px ☐ IV fluids (30cc/kg) ☐ Broad spectrum IV Abx ☐ Blood work/blood cultures ☐ Cap glucose ☐ Monitors ☐ O ₂	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> -O ₂ applied → Sat 96% - <u>Triggers</u> <i>For progression to next state</i> -No fluids/Abx by 3 min → Phase 2 -All actions complete → Phase 3	
2. Worsening HR: 130 BP: 80/40 O ₂ SAT:92 % GCS: 10	Becomes drowsy	<u>Expected Learner Actions</u> ☐ All actions above ☐ Call for help (ICU/RACE)	<u>Modifiers</u> - <u>Triggers</u> -No fluids/Abx by 6 min → PEA Arrest -All actions complete → Phase 3	
3. Initial Response HR: 110 BP: 104/55 O ₂ SAT:96 % GCS: 14		<u>Expected Learner Actions</u> ☐ Check BW ☐ Suspect cholangitis ☐ Call consultant for ERCP ☐ Call for help (ICU/RACE)	<u>Modifiers</u> - <u>Triggers</u> -All actions complete → Phase 4	
4. Disposition HR: 110 BP: 110/60 GCS: 14		<u>Expected Learner Actions</u> ☐ Recognize need for ongoing monitoring ☐ Arrange ERCP ☐ Update attending/senior resident	<u>Modifiers</u> - - <u>Triggers</u> -Speak with consultants → End of case	



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Appendix A: Laboratory Results

<u>CBC</u> WBC 17 Hgb 123 Plt 389 <u>Lytes</u> Na 138 K 3.8 Cl 96 Urea 7 Cr 105 Glucose 6.8 <u>VBG</u> pH 7.28 pCO ₂ 28 pO ₂ 68 HCO ₃ 28 Lactate 3.9	<u>Biliary</u> AST 60 ALT 40 GGT 540 ALP 330 Bili 60 Lipase 24 <u>Other</u> B-HCG negative
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Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!

No imaging or ultrasound required.

- Xrays will be normal.
- Ultrasound unavailable overnight.
- EKG shows sinus tach.



Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

1. Review the initial approach to sepsis
2. Discuss the approach to the patient with cholangitis and the importance/imperative of source control in the septic patient
3. Tools available to help communicate the case over the phone to the RACE physician/consultant

References

- 1.
- 2.
- 3.